

Maini Renewables — Site Assessment Report

Generated for Location: 26.489035, 80.254084 | Target Recipient: kpiyush032005@gmail.com

1. Multi-Height Wind Velocity Profiles

Using meteorological baseline records and Hellman Power Law shear adjustments, projected average wind speeds across varied vertical heights are outlined below:

Height Level	Projected Mean Wind Speed	Suitability Status
10 Meters	0.86 m/s	Low Kinetic Energy
25 Meters	0.98 m/s	Low Kinetic Energy
50 Meters	1.08 m/s	Low Kinetic Energy
100 Meters	1.19 m/s	Low Kinetic Energy

2. Vertical Axis Wind Turbine (VAWT) Suitability Analysis

Assessment Result: MARGINAL: Low average velocity profiles detected. Supplementary micro-wind channel orientation advised.

Vertical-axis configurations excel in turbulent micro-climates and low-speed zones. Based on the local spatial coordinates provided, the structural deployment profile matches operational parameters.